

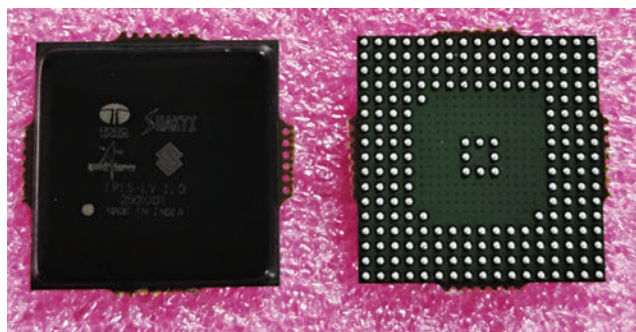
ture standpoint the challenges are minimal, but on the backend VLSI and physical design side, these are the main technical cliffs we need to overcome,” Patanjali highlights.

These are the kinds of constraints that remind us semiconductor progress is governed as much by physics as it is by global geopolitics. Advanced nodes require not just capital and policy alignment, but deep process expertise and materials innovation.

Building the next generation of hardware talent

Perhaps the most optimistic part of my conversation is about talent. India has spent decades becoming synonymous with software; now it must build a similar reputation in hardware. “That won’t happen overnight,” Patanjali says, while spelling out how things are picking up steam.

“The responsibility of the Shakti group is to break misconceptions with every tape-out. As we keep proving that complex processors can be built here, we see a new generation waking up and taking hardware design more seriously,” Patanjali says proudly. Each successful tape-out, each startup spun out of research labs, each commercial deployment chips away at the perception that cutting-edge hardware can only be built elsewhere.



For both professors, the ultimate measure of success is not policy announcements or research papers – it’s products.

“Five years from now, I want to see Shakti in applications in the field,” Gopal says. Nodding in agreement, Patanjali expands upon it, “You should be able to go and purchase a drone built using our processor or visit an organization and see a product powered by Shakti. Success for us is not code sitting in a repository – it’s an actual product you can visibly see and interact.”

That vision – of Indian-designed processors embedded in everyday devices – captures the real endgame of India’s semiconductor journey. Not just sovereignty as an abstract goal, but sovereignty manifested in tangible products.

If the voices from IIT Madras’ Shakti lab are any indication, India’s semiconductor story is no longer just about aspiration. It’s about capability slowly, methodically, coming online – one tape-out, one trained engineer, and one indigenous processor at a time. **d**

MADE IN INDIA SEMICONDUCTOR CHIPS (2023-2025)

2023

Sahasra

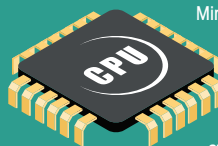
Sahasra Semiconductors became the first company to begin commercial production of “Made in India” memory products (like MicroSD cards and internal memory (NAND flash) packaging) at its facility in Bhiwadi, Rajasthan. Moving into advanced packaging for internal memory chips used in smartphones and laptops from 2025-onwards.



2024

Mindgrove “Secure IoT” S2401

IIT Madras-incubated startup Mindgrove Technologies launched India’s first commercial-grade high-performance SoC based on the RISC-V Shakti core (developed by IIT Madras). Built on a 28nm process, it was clocked at 700-MHz and designed to be 30% more cost-effective than global competitors, targeting smart meters, wearables, connected home appliances, and EV battery management systems.



2025

Vikram 32-bit Processor

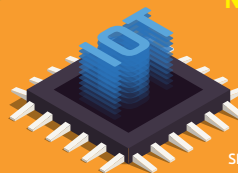
Unveiled at Semicon India 2025, the Vikram processor is a landmark achievement for India’s aerospace and defence sectors. Developed by ISRO’s Vikram Sarabhai Space Centre (VSSC) in collaboration with the Semiconductor Laboratory (SCL), Chandigarh, Vikram is a fully indigenous 32-bit microprocessor qualified for the extreme environments of space missions and rocket launches.



2025

Netrasemi Smart Vision Chips

Supported under the government’s Design Linked Incentive (DLI) scheme, Netrasemi made significant strides in 2025 with funding of 107 crore. The company is working on scaling production of AI-powered chips for smart vision, CCTV cameras, and Internet of Things (IoT) applications.



2025

Indigenous 7nm Server Processor

While still in the development/design phase, the Indian government announced a major leap into advanced nodes in late October 2025. Union Minister Ashwini Vaishnaw revealed the initiative to design a 7nm processor for high-performance computing for use in financial services, telecommunications, and strategic defence sectors.

